

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Department of Mathematics & Statistics

MTH 211: Theory of Statistics (Session: 2025-26, Semester: II)

Detailed Solutions for Problem Set 6

(Testing of Hypotheses - UMP test, MLR property)

Solution 1

Based on a random sample $X_1, \dots, X_n$, we derive the UMP size- α test for testing $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$. We first construct the Most Powerful (MP) size- α test for a simple null $H'_0 : \theta = \theta_0$ against a simple alternative $H'_A : \theta = \theta_1$ (where $\theta_1 > \theta_0$).

(a) $f_\theta(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0, \theta > 0$

By the Neyman-Pearson Lemma, the MP test is given by:

$$\phi(\underline{x}) = \begin{cases} 1 & \text{if } \frac{f_{\theta_1}(\underline{x})}{f_{\theta_0}(\underline{x})} \geq k \\ 0 & \text{otherwise} \end{cases}$$

The likelihood ratio is:

$$\frac{f_{\theta_1}(\underline{x})}{f_{\theta_0}(\underline{x})} = \left(\frac{\theta_0}{\theta_1}\right)^n \exp \left[\left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) \sum_{i=1}^n x_i \right] \geq k$$

Since $\theta_1 > \theta_0 \implies \left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) > 0$, the inequality reduces to $\sum_{i=1}^n x_i \geq c$. Thus, the test is:

$$\phi(\underline{x}) = \begin{cases} 1 & \text{if } \sum x_i \geq c \\ 0 & \text{otherwise} \end{cases}$$

To find c , we use the size condition $E_{\theta_0}[\phi(\underline{X})] = P_{\theta_0}(\sum X_i \geq c) = \alpha$. Under H_0 , $X_i \sim \text{Exp}(1/\theta_0)$, so $\sum X_i \sim \text{Gamma}(n, \theta_0)$. Therefore, $\frac{2\sum X_i}{\theta_0} \sim \chi_{2n}^2$.

$$P_{\theta_0} \left(\frac{2\sum X_i}{\theta_0} \geq \frac{2c}{\theta_0} \right) = \alpha \implies \frac{2c}{\theta_0} = \chi_{2n,\alpha}^2 \implies c = \frac{\theta_0}{2} \chi_{2n,\alpha}^2$$

Since c does not depend on θ_1 , this is the UMP(α) test for $H_0 : \theta = \theta_0$ vs $H_A : \theta > \theta_0$. Next, consider the power function of the test:

$$\beta_\phi(\theta) = P_\theta \left(\sum X_i > \frac{\theta_0}{2} \chi_{2n,\alpha}^2 \right) = P_\theta \left(\frac{2\sum X_i}{\theta} > \frac{\theta_0}{\theta} \chi_{2n,\alpha}^2 \right)$$

Since $\frac{2\sum X_i}{\theta} \sim \chi_{2n}^2$, the probability $P(\chi_{2n}^2 > \frac{\theta_0}{\theta} \chi_{2n,\alpha}^2)$ is monotonically increasing in θ . Thus, for $\theta \leq \theta_0$, $\beta_\phi(\theta) \leq \beta_\phi(\theta_0) = \alpha$. Therefore, ϕ is the UMP(α) test for $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$.

(c) $f_\theta(x) = \frac{\theta}{x^{\theta+1}}, \quad x \geq 1, \theta > 0$

The likelihood ratio for $\theta_1 > \theta_0$ is:

$$\frac{f_{\theta_1}(\underline{x})}{f_{\theta_0}(\underline{x})} = \left(\frac{\theta_1}{\theta_0}\right)^n \left(\prod_{i=1}^n x_i\right)^{-(\theta_1-\theta_0)} \geq k \implies \sum_{i=1}^n \log x_i \leq c$$

Let $Y = \log X$. Then $Y \sim \text{Gamma}(1, 1/\theta)$. Under H_0 , $\sum Y_i \sim \text{Gamma}(n, 1/\theta_0)$, which means $2\theta_0 \sum \log X_i \sim \chi_{2n}^2$. Using the size condition $P_{\theta_0}(\sum \log X_i \leq c) = \alpha$, we find $c = \frac{1}{2\theta_0} \chi_{2n, 1-\alpha}^2$. The power function is $\beta_\phi(\theta) = P_\theta \left(2\theta \sum \log X_i \leq \frac{\theta}{\theta_0} \chi_{2n, 1-\alpha}^2 \right)$. Since this is increasing in θ , it guarantees $\beta_\phi(\theta) \leq \alpha$ for all $\theta \leq \theta_0$. It is UMP.

$$(d) f_\theta(x) = (\sqrt{2\pi}\theta)^{-1} e^{-x^2/2\theta^2}, \quad -\infty < x < \infty, \theta > 0$$

The likelihood ratio for $\theta_1 > \theta_0$ gives:

$$\left(\frac{\theta_0}{\theta_1}\right)^n \exp \left[\frac{1}{2} \sum x_i^2 \left(\frac{1}{\theta_0^2} - \frac{1}{\theta_1^2} \right) \right] \geq k \implies \sum x_i^2 \geq c$$

Under H_0 , $\frac{\sum X_i^2}{\theta_0^2} \sim \chi_n^2$. Setting the size to α :

$$P_{\theta_0} \left(\frac{\sum X_i^2}{\theta_0^2} \geq \frac{c}{\theta_0^2} \right) = \alpha \implies c = \theta_0^2 \chi_{n, \alpha}^2$$

The power function $\beta_\phi(\theta) = P \left(\chi_n^2 > \frac{\theta^2}{\theta_0^2} \chi_{n, \alpha}^2 \right)$ is strictly increasing in θ . Thus, it is UMP for $H_0 : \theta \leq \theta_0$.

Solution 2

We need a UMP size- α test for $H_0 : \theta \geq \theta_0$ against $H_1 : \theta < \theta_0$ for $f_\theta(x) = \theta x^{\theta-1}$ where $0 < x < 1$. Let $\theta_1 < \theta_0$. The likelihood ratio is:

$$\frac{f_{\theta_1}(\underline{x})}{f_{\theta_0}(\underline{x})} = \left(\frac{\theta_1}{\theta_0}\right)^n \left(\prod_{i=1}^n x_i \right)^{\theta_1 - \theta_0} \geq k$$

Since $(\theta_1 - \theta_0) < 0$, taking the logarithm and rearranging gives $\sum (-\log x_i) \geq c$. Under H_0 , $Y_i = -\log X_i \sim \text{Exp}(1/\theta_0)$, so $2\theta_0 \sum (-\log X_i) \sim \chi_{2n}^2$. The size condition requires $c = \frac{1}{2\theta_0} \chi_{2n, \alpha}^2$. The power function $\beta_\phi(\theta) = P_\theta \left(2\theta \sum (-\log X_i) > \frac{\theta}{\theta_0} \chi_{2n, \alpha}^2 \right)$ is strictly decreasing in θ . Thus $\beta_\phi(\theta) \leq \alpha$ for all $\theta \geq \theta_0$, proving it is UMP.

Solution 3

Based on a single observation $X \sim \text{Poisson}(\theta)$, we want to derive the UMP size- α test for $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$.

First, consider the simple null $H_0' : \theta = \theta_0$ against the simple alternative $H_1' : \theta = \theta_1$ (where $\theta_1 > \theta_0$). The likelihood ratio is:

$$\lambda(x) = \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = \frac{e^{-\theta_1} \theta_1^x / x!}{e^{-\theta_0} \theta_0^x / x!} = e^{-(\theta_1 - \theta_0)} \left(\frac{\theta_1}{\theta_0} \right)^x$$

Since $\theta_1 > \theta_0$, the base $(\theta_1/\theta_0) > 1$. Hence, the likelihood ratio is monotonically increasing in x .

The Most Powerful (MP) size- α test rejects H_0 for large values of x . Since the Poisson distribution is discrete, we must use a randomized test to achieve exactly size α :

$$\phi(x) = \begin{cases} 1 & \text{if } x > c \\ \gamma & \text{if } x = c \\ 0 & \text{if } x < c \end{cases}$$

where the critical value c (an integer) and the randomization probability $\gamma \in [0, 1)$ are chosen such that the size condition holds:

$$E_{\theta_0}[\phi(X)] = P_{\theta_0}(X > c) + \gamma P_{\theta_0}(X = c) = \alpha$$

Since this test does not depend on the specific value of θ_1 , it is UMP for testing $H'_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$.

To show it is UMP for the composite null $H_0 : \theta \leq \theta_0$, we must analyze its power function and show it is monotonically increasing in θ :

$$\beta_\phi(\theta) = E_\theta[\phi(X)] = \sum_{x=c+1}^{\infty} f_\theta(x) + \gamma f_\theta(c)$$

We can rewrite this expression by strategically splitting the γ term:

$$\beta_\phi(\theta) = (1 - \gamma) \sum_{x=c+1}^{\infty} f_\theta(x) + \gamma \left(\sum_{x=c+1}^{\infty} f_\theta(x) + f_\theta(c) \right)$$

$$\beta_\phi(\theta) = (1 - \gamma) \sum_{x=c+1}^{\infty} f_\theta(x) + \gamma \sum_{x=c}^{\infty} f_\theta(x)$$

A known property relates the right tail of the Poisson distribution to the incomplete gamma function. Specifically:

$$\sum_{k=x}^{\infty} \frac{e^{-\theta} \theta^k}{k!} = \frac{\int_0^\theta e^{-t} t^{x-1} dt}{\int_0^\infty e^{-t} t^{x-1} dt} = I(\theta, x)$$

Using this identity, we can substitute the sums in the power function:

$$\beta_\phi(\theta) = (1 - \gamma)I(\theta, c + 1) + \gamma I(\theta, c)$$

Since the integrand $e^{-t} t^{x-1}$ is strictly positive for $t > 0$, the integral $\int_0^\theta e^{-t} t^{x-1} dt$ is a strictly increasing function of its upper limit θ . Therefore, both $I(\theta, c + 1)$ and $I(\theta, c)$ are strictly increasing functions of θ .

Because $\gamma \in [0, 1)$, $\beta_\phi(\theta)$ is a convex combination of strictly increasing functions, making $\beta_\phi(\theta)$ strictly increasing in θ . This monotonic property implies that for all $\theta \leq \theta_0$:

$$\beta_\phi(\theta) \leq \beta_\phi(\theta_0) = \alpha$$

Thus, the size condition is satisfied for all $\theta \in H_0$, proving that $\phi(x)$ is the UMP size- α test for $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$.

Solution 4

Verification of the Monotone Likelihood Ratio (MLR) property.

(a) $f_\theta(x) = 1$ for $\theta < x < \theta + 1$

Let $\theta_1 < \theta_2$.

$$\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} = \begin{cases} 0 & \text{if } \theta_1 < x \leq \theta_2 \\ 1 & \text{if } \theta_2 < x < \theta_1 + 1 \\ \infty & \text{if } \theta_1 + 1 \leq x < \theta_2 + 1 \end{cases}$$

Since the ratio goes from $0 \rightarrow 1 \rightarrow \infty$ as x increases, it is a non-decreasing function of x . Thus, the family **has MLR** in $T(X) = X$.

(b) Cauchy Distribution $f_\theta(x) = \frac{1}{\pi(1+(x-\theta)^2)}$

For $\theta_1 < \theta_2$:

$$\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} = \frac{1 + (x - \theta_1)^2}{1 + (x - \theta_2)^2}$$

As $x \rightarrow \pm\infty$, this ratio approaches 1. The function rises to a peak and falls, meaning it is not monotonic. The family **does not have MLR** in X .

(c) Exponential Distribution

This is a standard one-parameter exponential family. The ratio is strictly increasing in x . It **has MLR** in $T(X) = X$.

Solution 5

Random sample $X_1, \dots, X_n \sim U(0, \theta)$. Test $H_0 : \theta \leq \theta_0$ vs $H_A : \theta > \theta_0$. The family has MLR in the sufficient statistic $T(X_n) = X_{(n)}$. The standard UMP test rejects for $X_{(n)} \geq c$.

$$P_{\theta_0}(X_{(n)} \geq c) = \alpha \implies 1 - \left(\frac{c}{\theta_0}\right)^n = \alpha \implies c = \theta_0(1 - \alpha)^{1/n}$$

Power function: $\beta_\phi(\theta) = 1 - (1 - \alpha) \left(\frac{\theta_0}{\theta}\right)^n$.

Now consider an alternative test ϕ^* :

$$\phi^*(\underline{x}) = \begin{cases} 1 & \text{if } x_{(n)} > \theta_0 \\ \alpha & \text{if } x_{(n)} \leq \theta_0 \end{cases}$$

Size: $E_{\theta_0}[\phi^*(\underline{x})] = \alpha P_{\theta_0}(X_{(n)} \leq \theta_0) = \alpha(1) = \alpha$. Power for $\theta > \theta_0$:

$$\beta_{\phi^*}(\theta) = P_\theta(X_{(n)} > \theta_0) + \alpha P_\theta(X_{(n)} \leq \theta_0) = \left[1 - \left(\frac{\theta_0}{\theta}\right)^n\right] + \alpha \left(\frac{\theta_0}{\theta}\right)^n = 1 - (1 - \alpha) \left(\frac{\theta_0}{\theta}\right)^n$$

Since $\beta_\phi(\theta) = \beta_{\phi^*}(\theta)$ for all $\theta > \theta_0$, both are UMP size- α tests. Thus, the UMP test is **not unique**.